

Data leak worksheet

Incident summary: A sales manager shared access to a folder of internal-only documents with their team during a meeting. The folder contained files associated with a new product that has not been publicly announced. It also included customer analytics and promotional materials. After the meeting, the manager did not revoke access to the internal folder, but warned the team to wait for approval before sharing the promotional materials with others.

During a video call with a business partner, a member of the sales team forgot the warning from their manager. The sales representative intended to share a link to the promotional materials so that the business partner could circulate the materials to their customers. However, the sales representative accidentally shared a link to the internal folder instead. Later, the business partner posted the link on their company's social media page assuming that it was the promotional materials.

Control	Least privilege
Issue(s)	<i>What factors contributed to the information leak?</i> <i>The data leak was primarily caused by a failure to revoke access permissions after a specific business need was met, combined with human error during information sharing. The manager’s oversight in leaving the folder accessible allowed a representative to accidentally share internal-only analytics rather than just the intended promotional materials. Significant misuse of data in transit.</i>
Review	<i>What does NIST SP 800-53: AC-6 address?</i> <i>AC-6 defines the principle of least privilege, which mandates that users should only be granted the minimum level of access necessary to perform their specific job functions. This framework is designed to be customizable,</i>

	<i>helping businesses prevent users from operating at privilege levels higher than required for their objectives.</i>
Recommendation(s)	<i>How might the principle of least privilege be improved at the company?</i> <i>Based on the NIST SP 800-53: AC-6 control enhancements, the following improvements should be implemented:</i> 1. Restrict access based on user role: <i>Ensure sales representatives only have access to "External" folders, while "Internal" analytics remain restricted to management.</i> 2. Automatically revoke access: <i>Implement a policy where access to sensitive internal folders expires automatically after a set period, such as the conclusion of a meeting.</i>
Justification	<i>How might these improvements address the issues?</i> <i>Implementing these enhancements will reduce the likelihood of future leaks by removing the "human error" factor of forgetting to unshare files. By automating access expiration and enforcing role-based restrictions, the company ensures that even if a link is shared accidentally, unauthorized external parties will be blocked by the system's underlying permission structure.</i>

Security plan snapshot

The NIST Cybersecurity Framework (CSF) uses a hierarchical, tree-like structure to organize information. From left to right, it describes a broad security function, then becomes more specific as it branches out to a category, subcategory, and individual security controls.

Function	Category	Subcategory	Reference(s)
Protect	PR.DS: <i>Data security</i>	PR.DS-5: <i>Protections against data leaks.</i>	NIST SP 800-53: AC-6

In this example, the implemented controls that are used by the manufacturer to protect against data leaks are defined in NIST SP 800-53—a set of guidelines for securing the privacy of information systems.

Note: References are commonly hyperlinked to the guidelines or regulations they relate to. This makes it easy to learn more about how a particular control should be implemented. It's common to find multiple links to different sources in the references columns.

NIST SP 800-53: AC-6

NIST developed SP 800-53 to provide businesses with a customizable information privacy plan. It's a comprehensive resource that describes a wide range of control categories. Each control provides a few key pieces of information:

- **Control:** A definition of the security control.
- **Discussion:** A description of how the control should be implemented.
- **Control enhancements:** A list of suggestions to improve the effectiveness of the control.

AC-6	Least Privilege
	Control: Only the minimal access and authorization required to complete a task or function should be provided to users.
	Discussion: Processes, user accounts, and roles should be enforced as necessary to achieve least privilege. The intention is to prevent a user from operating at privilege levels higher than what is necessary to accomplish business objectives.
	Control enhancements: ● Restrict access to sensitive resources based on user role. ● Automatically revoke access to information after a period of time. ● Keep activity logs of provisioned user accounts. ● Regularly audit user privileges.

Note: In the category of access controls, SP 800-53 lists least privilege sixth, i.e. AC-6.